Week Lab Manual: Computer Abstractions & Performance

1-Introduction

This lab manual provides a detailed overview and practical exercises for understanding computer performance metrics, focusing on the CPU Performance Equation, MIPS (Millions of Instructions Per Second), CPI (Clock Cycles Per Instruction), and Amdahl's Law. These concepts are fundamental to evaluating and optimizing computer system performance. This manual aligns with the topics covered in Stallings: Chapter [ ].

Learning Objectives: Upon completion of this lab, students will be able to:

- Understand the components of the CPU Performance Equation.
- Calculate CPU execution time, MIPS, and CPI for given scenarios.
- Apply Amdahl's Law to predict performance improvements from system enhancements.
- Analyze the impact of various architectural decisions on overall system performance

2-The CPU Performance Equation

Computer performance is often measured by the time it takes to execute a program. The CPU execution time for a program can be broken down into three key components

Instruction Count (IC): The total number of instructions executed by the program. This is influenced by the Instruction Set Architecture (ISA) and the compiler's efficiency.

Clock Cycles Per Instruction (CPI): The average number of clock cycles required to execute a single instruction. This depends on the CPU's organization (e.g., pipelining, superscalar architecture) and the mix of instructions in the program.

Clock Cycle Time (T_c): The duration of a single clock cycle, typically measured in seconds. It is the inverse of the clock rate (frequency).

Alternatively, if the clock rate (Frequency, F) is known:

$$\text{CPU Time} = \frac{\text{Instruction Count (IC)} \times \text{Clock Cycles Per Instruction (CPI)}}{\text{Clock Rate (F)}}$$

Solved Example 2.1: Calculating CPU Time

Problem: Consider a program that executes 100 billion instructions (IC = 100×10^9). The processor has an average CPI of 1.5 and a clock rate of 3 GHz. Calculate the CPU execution time for this program.

Solution: Given:

- IC = 100×10^9 instructions
- CPI = 1.5 cycles/instruction
- Clock Rate (F) = 3 GHz = 3×10^9 cycles/second

Using the formula:

$$\text{CPU Time} = \frac{\text{IC} \times \text{CPI}}{\text{F}}$$

$$\text{CPU Time} = \frac{(100 \times 10^9) \times 1.5}{3 \times 10^9}$$

$$\text{CPU Time} = \frac{150 \times 10^9}{3 \times 10^9}$$

$$\text{CPU Time} = 50 \text{ seconds}$$

Answer: The CPU execution time for the program is 50 seconds.

Example1: Consider the execution of a program with 2 million instructions on 400 MH processor, program consists of four major types ALU with 60 of instructions, Load with 18, memory reference 10 , and branch with 12 in which each has number of cycles respectively 1,2,8,4 find the performance

Op	Freq	(% Time)	Cycles	CPI(i)
ALU	60	(0.6)	1	1x.6=0.6
Load	18	(0.18)	2	2x0.18=.36
Store	10	(0.1)	8	8x0.1=0.8
Branch	12	(0.12)	4	4x0.12=0.48
	100	Weight=1		$\sum CPI_i = 2.24$

Typical Mix

3. MIPS (Millions of Instructions Per Second)

MIPS is a measure of processor speed, indicating how many millions of instructions a processor can execute per second. It is often used to compare the performance of different processors, though it can be misleading if not used carefully (e.g., different ISAs or instruction mixes).

Formula:

$$\text{MIPS} = \frac{\text{Instruction Count (IC)}}{\text{CPU Time} \times 10^6} = \frac{\text{Clock Rate (F)}}{\text{CPI} \times 10^6}$$

Solved Example 3.1: Calculating MIPS

Problem: Using the data from Solved Example 2.1 (IC = 100×10^9, CPI = 1.5, F = 3 GHz, CPU Time = 50 seconds), calculate the MIPS rating of the processor.

Solution: Using the formula:

$$\text{MIPS} = \frac{\text{IC}}{\text{CPU Time} \times 10^6}$$

$$\text{MIPS} = \frac{100 \times 10^9}{50 \times 10^6}$$

$$\text{MIPS} = \frac{100000}{50}$$

$$\text{MIPS} = 2000 \text{ MIPS}$$

Alternatively, using the clock rate and CPI:

$$\text{MIPS} = \frac{\text{F}}{\text{CPI} \times 10^6}$$

$$\text{MIPS} = \frac{3 \times 10^9}{1.5 \times 10^6}$$

$$\text{MIPS} = \frac{3000}{1.5}$$

$$\text{MIPS} = 2000 \text{ MIPS}$$

Answer: The processor has a MIPS rating of 2000.

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4. Amdahl's Law

Amdahl's Law is a formula that gives the theoretical speedup in latency of the execution of a task at fixed workload, that can be expected of a system whose resources are improved. It is often used in parallel computing to predict the maximum speedup when only a portion of the task can be parallelized.

Formula:

$$\text{Speedup} = \frac{1}{(1-f) + \frac{f}{s}}$$

Where:

- **Speedup:** The overall performance improvement factor.
- **f:** The fraction of the execution time that can be enhanced or improved (e.g., the fraction of a program that can be parallelized).
- **s:** The speedup factor of the enhanced portion (e.g., how many times faster the parallelized part becomes).

Solved Example 4.1: Applying Amdahl's Law

Problem: Suppose a program runs in 100 seconds. Floating-point operations account for 20 seconds of the execution time. If we can make the floating-point unit 5 times faster, what is the overall speedup?

Solution: Given:

- Total execution time = 100 seconds
- Time spent on floating-point operations = 20 seconds
- Fraction of execution time that can be enhanced (f) = 20 / 100 = 0.2
- Speedup factor of the enhanced portion (s) = 5

Using Amdahl's Law:

$$\text{Speedup} = \frac{1}{(1-f) + \frac{f}{s}}$$

$$\text{Speedup} = \frac{1}{(1-0.2) + \frac{0.2}{5}}$$

$$\text{Speedup} = \frac{1}{0.8 + 0.04}$$

$$\text{Speedup} = \frac{1}{0.84}$$

$$\text{Speedup} \approx 1.19$$

Answer: The overall speedup of the program will be approximately 1.19 times.

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5. Practice Problems

1. **CPU Time Calculation:** A new processor has a clock rate of 4 GHz. A program executes 2×10^{10} instructions with an average CPI of 1.2. Calculate the CPU execution time for this program.

2. **MIPS Calculation:** For the program and processor in problem 1, what is the MIPS rating of the processor?

3. **Amdahl's Law Application:** A program spends 60% of its time performing integer operations. If we can make the integer unit 3 times faster, what is the overall speedup of the program?

4. **Combined Performance:** Processor A has a clock rate of 2.5 GHz and executes a program in 80 seconds with an average CPI of 1.8. Processor B has a clock rate of 3.0 GHz and executes the same program in 70 seconds with an average CPI of 1.6. Which processor is faster, and by what factor?

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6. References

[1] Stallings, W. (2019). *Computer Organization and Architecture: Designing for Performance* (11th ed.). Pearson. Chapter 2: Computer Abstractions and Technology.